

PROJECT AQUA-NEPTUNE: TECHNICAL SPECIFICATIONS

Engineering Architecture, Materials, and Fluid-Dynamics Manifesto

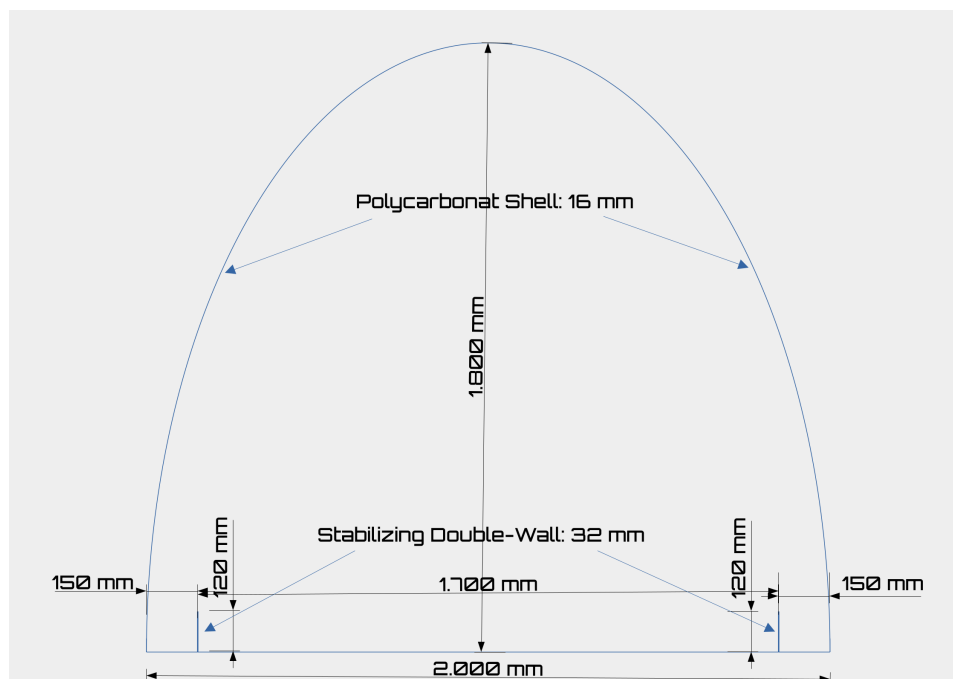
1. SYSTEM DIMENSIONS & ARCHITECTURAL GEOMETRY

The infrastructure is engineered as a **Two-Part Modular System** consisting of a rigid structural ground tray and a flexible, detached catenary roof structure.

Note: All associated graphics and repository images are to be considered a schematic representation / structural layout visualization only.

- **Total Exterior Width:** 2000 mm (2.0 Meters)
- **Maximum Apex Height:** 1800 mm (1.8 Meters)
 - *Thermodynamic Optimization:* This exact height minimizes the internal air volume to guarantee rapid solar air heating, while maintaining full human ergonomics for maintenance crews.
- **Aqueduct Longitudinal Gradient:** < 0.5% (0.2% - 0.4% target slope).
 - *Fluid Mechanics:* Fluid movement is driven exclusively by the Earth's gravity, completely eliminating electrical pumping power over long distances.

Cross-Section Design (Scale Layout)



Component Deconstruction (Ground Tray Profile)

- **Central Seawater Channel (Middle Tray):** 1700 mm wide, resting completely flat on the desert ground base level.
 - *Thermodynamic Rule:* Maximizes the surface-area-to-volume ratio of the seawater to exponentially accelerate solar evaporation.
 - **Lateral Catchment Channels (Left & Right Channel):** 150 mm wide, 120 mm deep, located completely **inside** the protective outer shell to avoid fluid loss.
 - *Surface Tension Rule:* Pure condensed water slides down the inner curved Polycarbonate roof via surface tension and drops directly into these integrated outer channels.
 - **Internal Stabilizing Joint Profile:** 120 mm height, built with a double-wall thickness (32 mm total structural width). This profile acts as a structural fold (**Folder Structure**), multiplying rigidity against lateral liquid force while safely acting as an anti-spill barrier between seawater and fresh distillate.
 - **Outer Shell Structural Flanges:** 120 mm height connection walls at the absolute left and right perimeter. These 120 mm walls are geometric-engineered with a slight inward curve (radial relief) to absorb high cyclic thermodynamic expansion and prevent structural material fatigue.
 - **Integrated Factory-Pre-Assembled Gasket System:** The inner vertical face of both 120 mm outer flanges comes equipped with a factory-bonded, heavy-duty 120 mm x 6 mm EPDM elastomeric profile gasket. This ensures absolute airtight and watertight sealing immediately upon sliding the roof module into place on site.
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2. MATERIAL & SELECTION SPECIFICATIONS

A. The Detached Catenary Roof (Catenary Vault)

- **Material:** Multi-Wall Hollow-Chamber-Polycarbonate (Twin-Wall or Triple-Wall sheets).
- **Wall Thickness:** 10 mm - 16 mm.
 - *Thermal Insulation:* The internal air gaps within the sheet trap thermal energy, ensuring evaporation continues autonomously for several hours after sunset.
- **Surface Coating:** Single-sided co-extruded UV-Protection (Facing Outwards). Preventative engineering against yellowing, structural embrittlement, and optical degradation from harsh desert solar radiation.

B. The Structural Ground Tray (Ground Tray Module)

- **Base Structure:** Heavy-duty, solid extruded Food-Grade Polycarbonate sheets.
 - **Sealing System:** A 120 mm wide, 5 mm - 6 mm thick heavy-duty elastomeric, non-toxic EPDM profile gasket line lined on the inner side of the outer structural flanges.
 - **Fastening System:** The detached roof element is slid into the gasket line and mechanically locked into the inner side of the channel structure utilizing industrial Automotive-Grade plastic expansion rivets, allowing stress-free thermal shifting.
 - **Central Channel Thermal Coating (Black):** Coated with a specialized, highly durable, food-grade certified black Epoxy or Polyurethane matrix (FDA compliant / EC 1935/2004). Guarantees maximum solar absorption (low albedo) while completely preventing toxic leaching into the NaCl salt harvest.
 - **Lateral Catchment Coating (Reflective):** The inner surfaces of the fresh H²O channels are coated with an ultra-pure, food-grade Titanium Dioxide (TiO₂) reflective white shield or medical-grade mirror-polished liners. This reflects incoming solar rays to keep the harvested freshwater cool and completely prevent heat-induced re-evaporation.
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3. THE THERMODYNAMIC SEPARATION CURTAIN

At the precise length where fractional evaporation finishes, a **Thermodynamic Curtain** seals off the channel.

- **Function:** It creates a complete air-mass barrier between the high-humidity evaporation zone and the bone-dry post-processing crystallization zone.
 - **Result:** It prevents water vapor from re-humidifying the precipitated salt.
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4. THE JET-BLACK CONVEYOR HARVESTER

- **The Anti-Albedo Problem:** Left static, crystalline salt creates a mirror effect (high albedo), reflecting solar rays and destroying the thermal efficiency of the greenhouse floor.
 - **The Solution:** A mechanical, continuous, ultra-slow conveyor belt running exclusively behind the thermodynamic curtain.
 - **Color:** Jet-Black (Maximum Solar Absorption).
 - **Speed Profile:** Operates at a micro-velocity (snail's pace) entirely inside the dry greenhouse zone until the **Pure NaCl (Salt)** is completely bone-dry for non-abrasive packaging into sacks or containers.
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5. VEHICLE POWERTRAIN INTEGRATION

The green hydrogen produced from the distilled water feeds a decentralized, closed-loop automotive network:

- **Engine Configuration:** Hydrogen Internal Combustion Engine (H₂-ICE) serving exclusively as a **Serial Hybrid Generator**.
 - **Operational Mode:** Fixed RPM (Thermodynamic Sweet Spot, e.g., locked at 2500 RPM).
 - **Efficiency Yield:** Eliminates throttle losses and transitional acceleration spikes, yielding a steady thermal efficiency of 45% - 50%.
 - **Buffer System:** Energy is pushed into a small, lightweight buffer battery. Heavy, resource-destructive 100 kWh chemical batteries are completely engineered out of the vehicle.
 - **Exhaust Condensation Catchment:** The vehicle tailpipe is equipped with an integrated condensation heat-exchanger. Pure H₂O exhaust emissions are liquefied, retained in an onboard tank, and deposited back at refueling stations.
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6. TOTAL PROJECT COMMODITY OUTPUTS

Project AQUA-NEPTUNE does not operate on artificial market scarcity. It continuously yields:

1. **Green Hydrogen (H₂):** 120 MJ/kg high-density energy stock.
 2. **Pure Distilled & Drinking Water (H₂O):** For human life and agricultural desert terraforming.
 3. **Food-Grade & Industrial Salt (NaCl):** Clean crystalline commodities.
 4. **Strategic Residual Minerals:** Precipitated extraction of Lithium, Magnesium, and Calcium.
 5. **Medical Oxygen (O₂):** High-purity electrolysis byproduct.
 6. **Baseload Solar Electricity:** Direct surplus solar PV generation routed into regional local grids.
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7. RE-SPLITTING MATRIX (THE TRANSNATIONAL ENERGY LOOP)

The clean water returned by vehicle drivers at destination countries is subjected to **Secondary Electrolysis** utilizing regional daytime Solar PV and Wind grids. This locks the energy into a secondary local cycle, transforming the water into an eternal transnational energy carrier.

8. INTELLECTUAL PROPERTY & GLOBAL LICENSE STATUS

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